

UMS_SYSTEM - Project Report

1. Project Overview

Key Features Implemented:

- **User Login System** with role-based access (Admin, Staff, Lecturer, Student).
- **Course and Subject Management** with linking between courses and subjects.
- **Exam Creation and Marks Entry**, allowing Admin and Lecturers to manage student assessments.
- **Attendance Tracking** with validation to prevent duplicate entries.
- **Timetable Scheduling**, assigning subjects, rooms, time slots, and lecturer to groups.
- **Change Password** functionality on first login and anytime via profile.
- **Dashboard with Panel-Based Dynamic Form Loading** for smooth navigation and better UX.
- .
- Technologies Used:
 - - C# (Windows Forms)
 - - SQLite database
 - - MVC design pattern
 - - Visual Studio IDE
 - - NuGet package: System.Data.SQLite

Challenges Faced and Solutions:

- **Challenge: Handling first-time Admin setup**

Solution: Used default Admin credentials and implemented a force password-change logic after first login.

- **Challenge: Secure and modular user management**

Solution: Separated forms for Students and Lecturers, and a unified form for Admin/Staff. Role-based filtering was used.

- **Challenge: Preventing duplicate attendance entries**

Solution: Applied a **UNIQUE constraint** in the database and added frontend validation to ensure only one entry per student per subject per day.

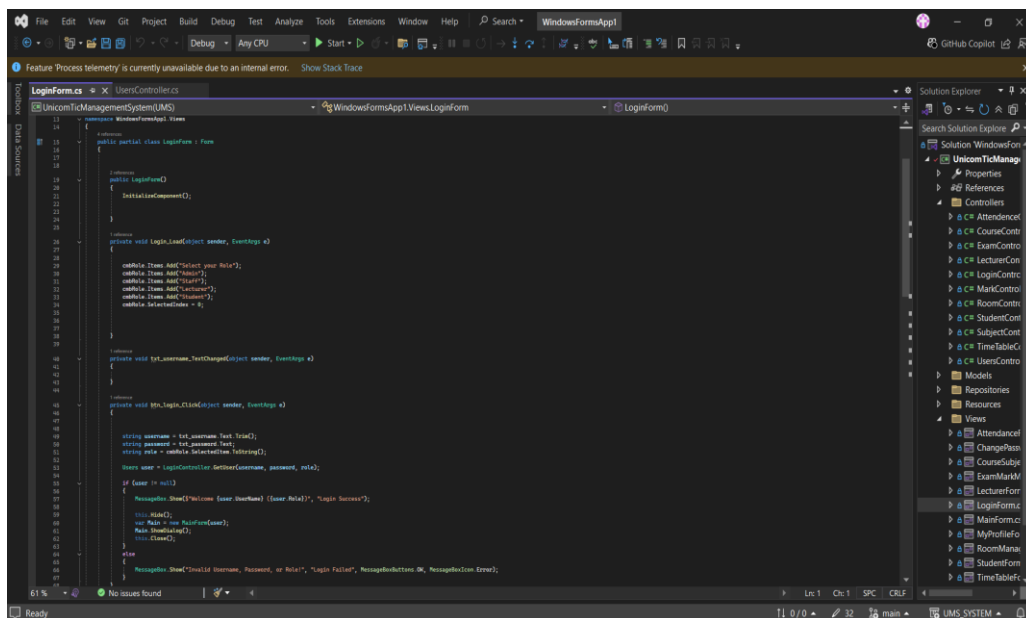
- **Challenge: Keeping UI clean and manageable**

Solution: Implemented a **panel-based dashboard** where child forms are loaded dynamically on menu or button clicks instead of opening multiple windows.

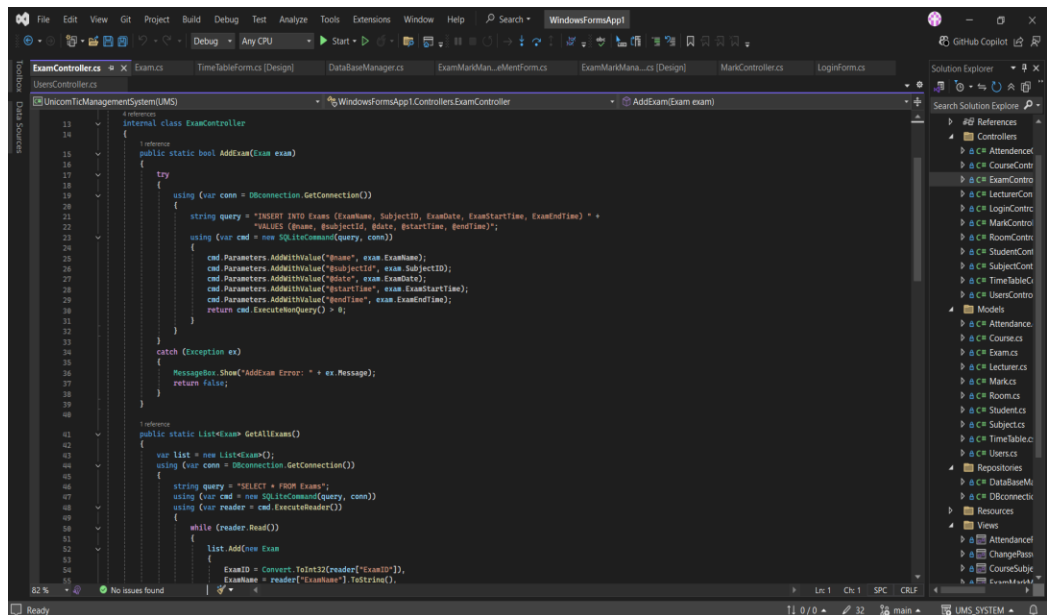
2. Code Samples (Screenshots)

1. User Login Logic

Role-based redirection after successful login

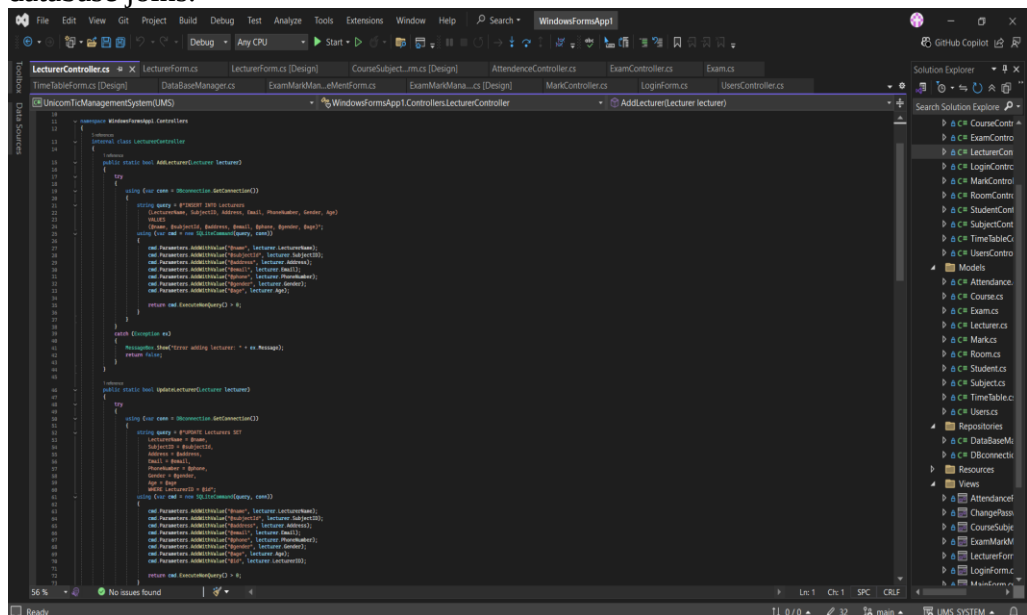


```
1  LoginForm0.cs
2  UnicomTicManagementSystem(UMS)
3  WindowsFormsApp1.Views.LoginForm0
4  LoginForm0
5
6  public partial class LoginForm : Form
7  {
8      // Fields
9      private string username;
10     private string password;
11     private string role;
12
13     // Methods
14     public LoginForm()
15     {
16         InitializeComponent();
17     }
18
19     // Login Logic
20     private void Login_Click(object sender, EventArgs e)
21     {
22         // Check if username and password are not empty
23         if (username == "" || password == "")
24         {
25             MessageBox.Show("Please enter username and password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
26             return;
27         }
28
29         // Check if username and password are correct
30         if (username == "admin" && password == "admin")
31         {
32             role = "Admin";
33         }
34         else if (username == "lecturer" && password == "lecturer")
35         {
36             role = "Lecturer";
37         }
38         else if (username == "student" && password == "student")
39         {
40             role = "Student";
41         }
42         else
43         {
44             MessageBox.Show("Invalid username or password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
45             return;
46         }
47
48         // Redirect to the appropriate form based on the role
49         if (role == "Admin")
50         {
51             goto Main;
52         }
53         else if (role == "Lecturer")
54         {
55             goto LecturerForm;
56         }
57         else if (role == "Student")
58         {
59             goto StudentForm;
60         }
61     }
62
63     // Password Change Logic
64     private void ChangePassword_Click(object sender, EventArgs e)
65     {
66         // Check if username and password are not empty
67         if (username == "" || password == "")
68         {
69             MessageBox.Show("Please enter username and password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
70             return;
71         }
72
73         // Check if password is correct
74         if (password == "admin" || password == "lecturer" || password == "student")
75         {
76             MessageBox.Show("Invalid password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
77             return;
78         }
79
80         // Change password
81         password = "newpassword";
82     }
83
84     // Role Change Logic
85     private void RoleChange_Click(object sender, EventArgs e)
86     {
87         // Check if username and password are not empty
88         if (username == "" || password == "")
89         {
90             MessageBox.Show("Please enter username and password.", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
91             return;
92         }
93
94         // Change role
95         role = "Admin";
96     }
97 }
```



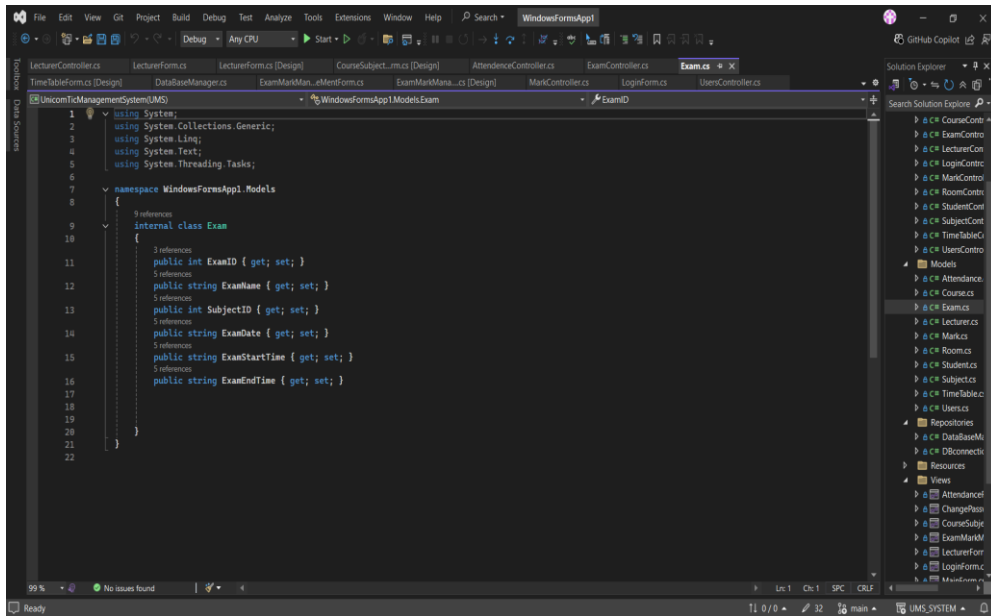
2. Subject Management

- Linking subjects with courses and lecturers using ComboBoxes and database joins.



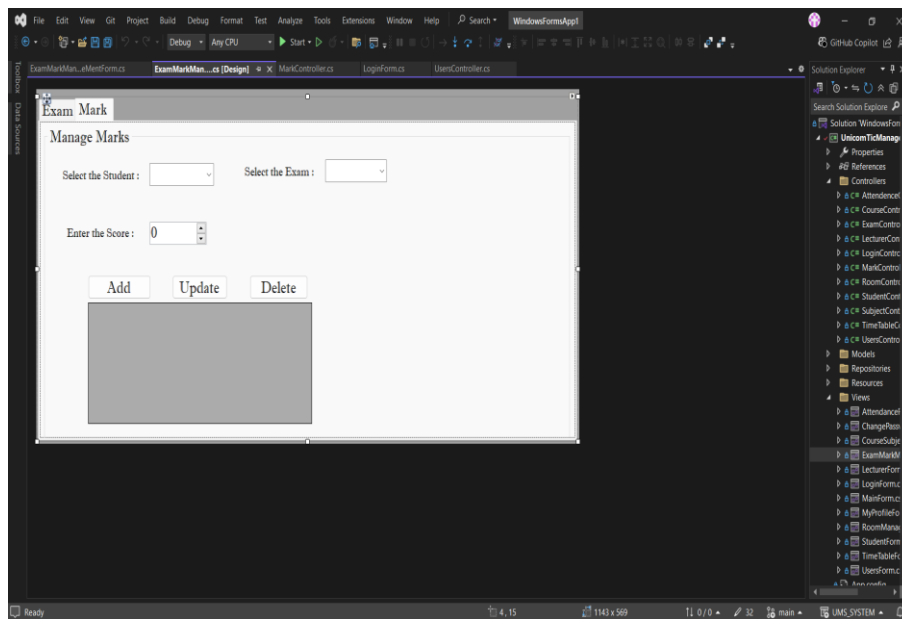
3. Exam Module

- Creating exams and linking to subjects with date & time selectors.



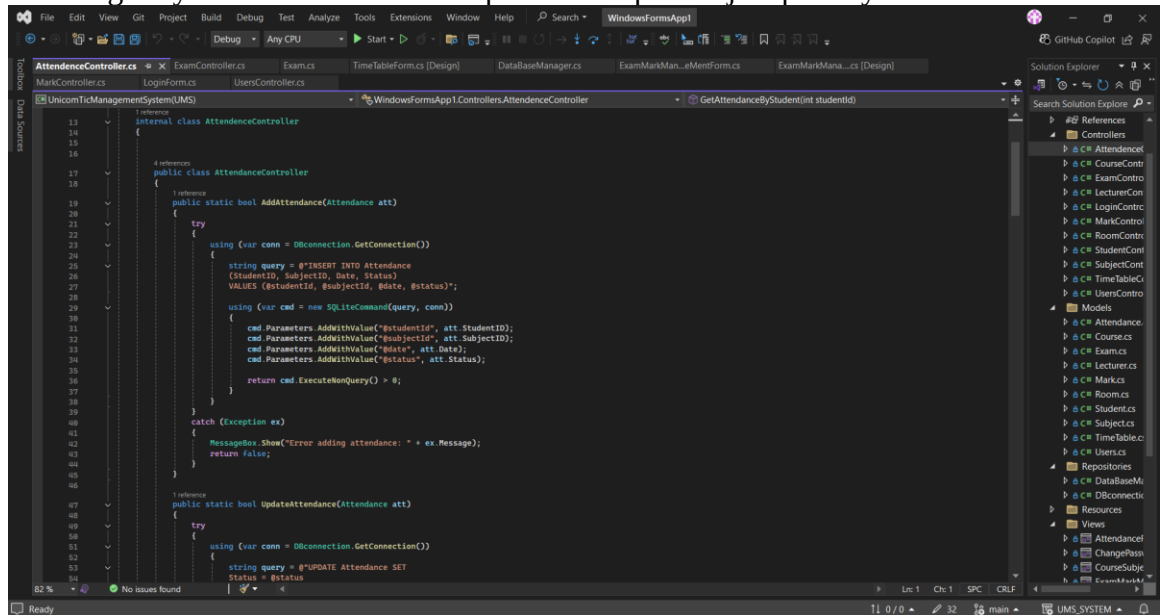
Marks Entry

3. Selecting exam and student, then entering scores (0–100) with validation.



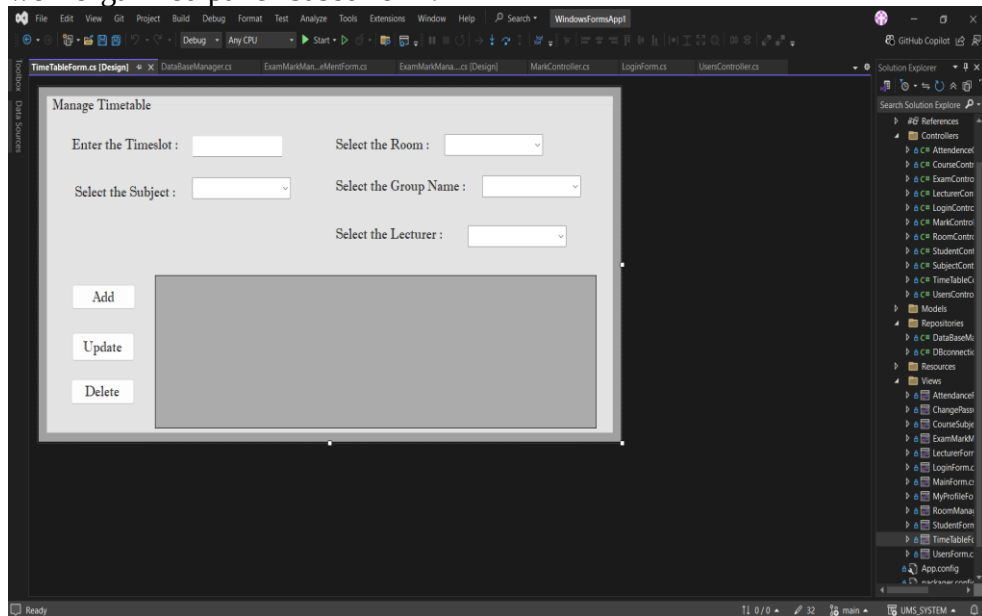
4. Attendance Validation

- Ensuring only one attendance record per student per subject per day.



5. Timetable Panel UI

- Assigning time slots to subjects, grouping them, and showing data in a well-organized panel-based form.



6. My profile

User searching the personal details

